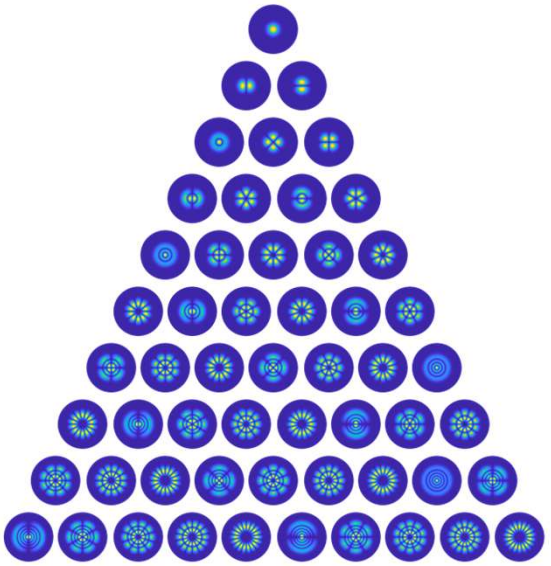
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

Lecture 6



Prof. Dr.-Ing. Georg Rademacher

Linearly polarized (LP) waves

University of Stuttgart
Germany

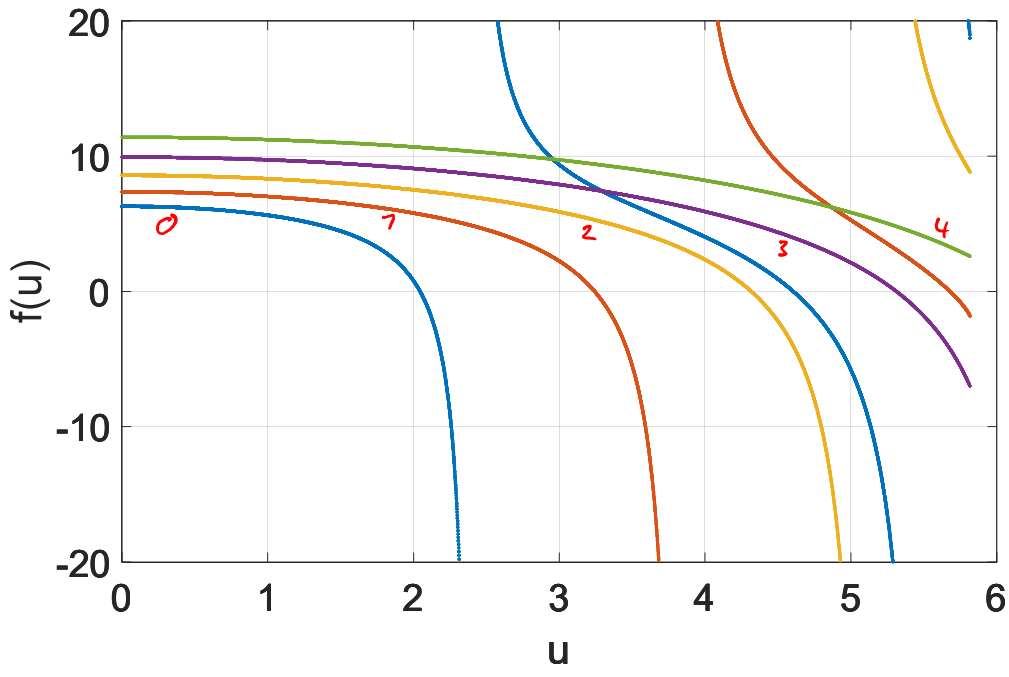
$\lambda = 1{,}55\,\mu m$, $a = 7\,\mu m$, $NA = 0.205 \rightarrow V = 5.817$

$V = k_0 \cdot a \cdot NA$

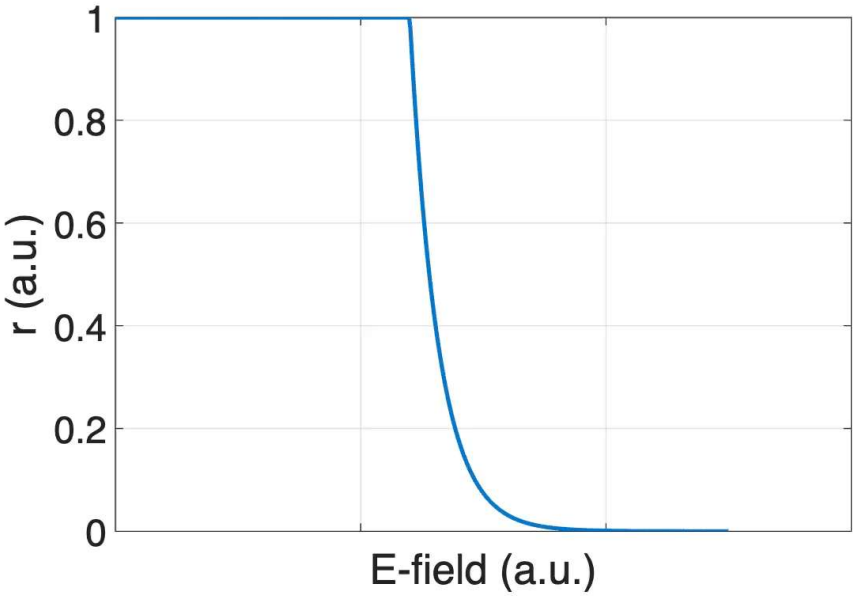
$V^2 = u^2 + v^2$

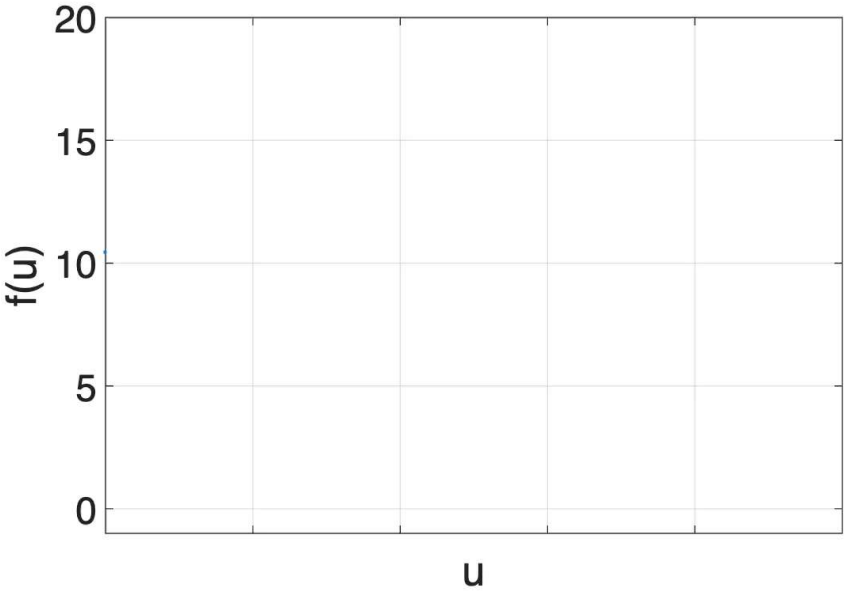
$u = V\sqrt{1 - B}$

LP_{lp}



Linearly polarized (LP) waves



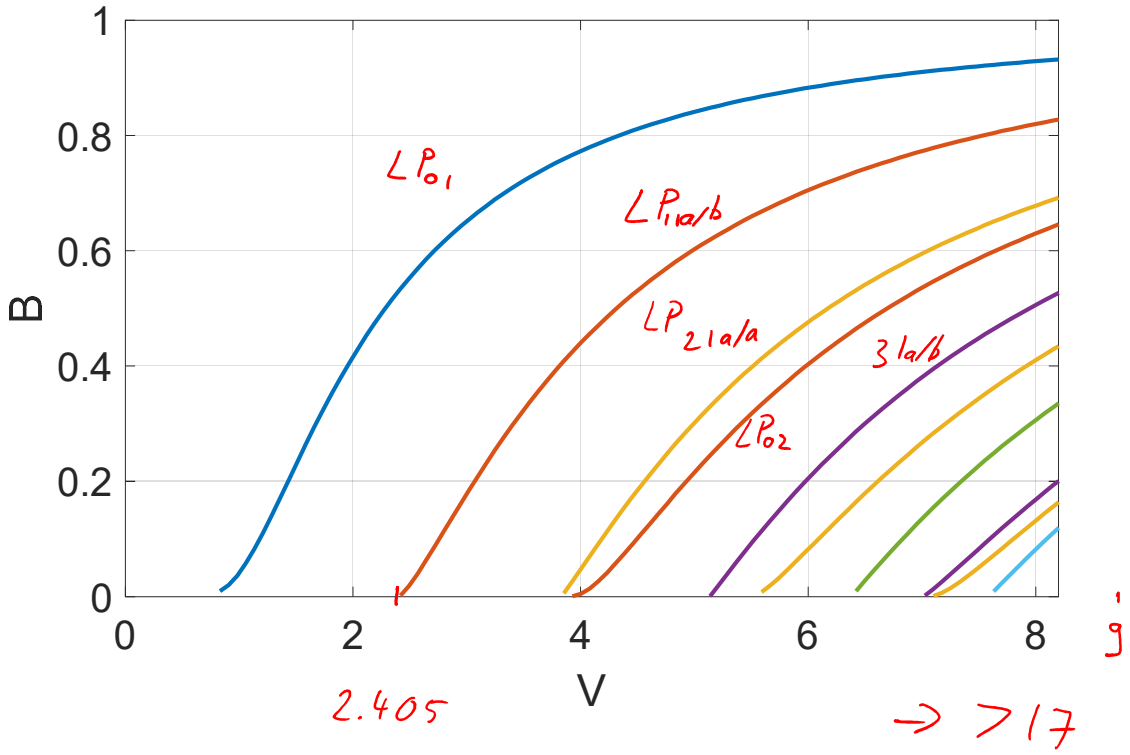


Linearly polarized (LP) waves

University of Stuttgart
Germany

$V = k_0 \cdot a \cdot NA$

$B \approx \frac{\frac{\beta}{k_0} - n_2}{n_1 - n_2}$



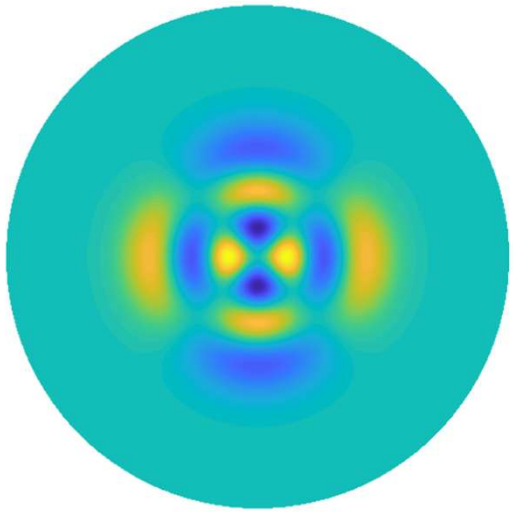


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Germany

Concept Test

University of Stuttgart
Germany

Which Linearly polarized (LP) mode is shown?



A) LP_{23}

B) LP_{32}

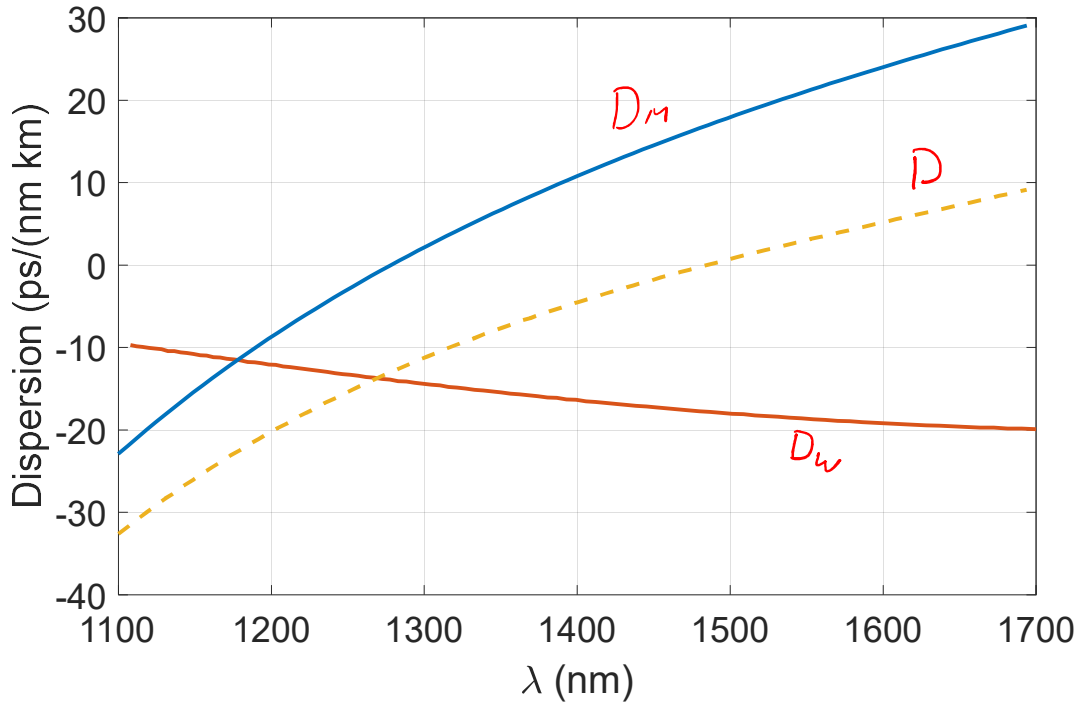
C) LP_{42}

https://ilias3.uni-stuttgart.de/vote/K2XW

Chromatic dispersion of the fundamental mode

University of Stuttgart
Germany

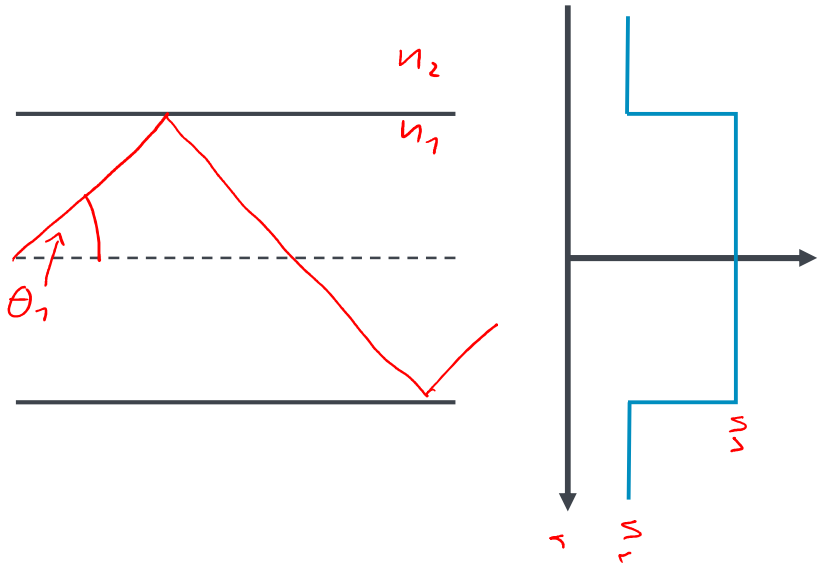
$a = 2.4\ \mu m$, $NA = 0.15$

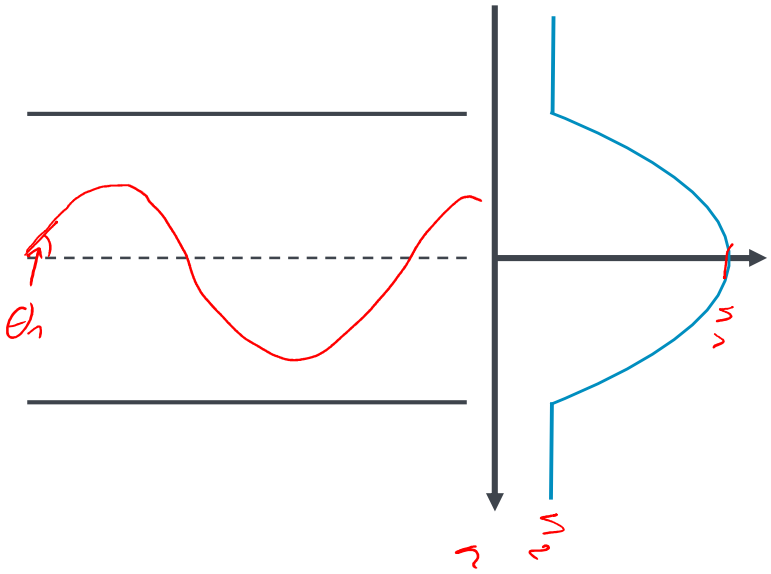


Graded-index fibers

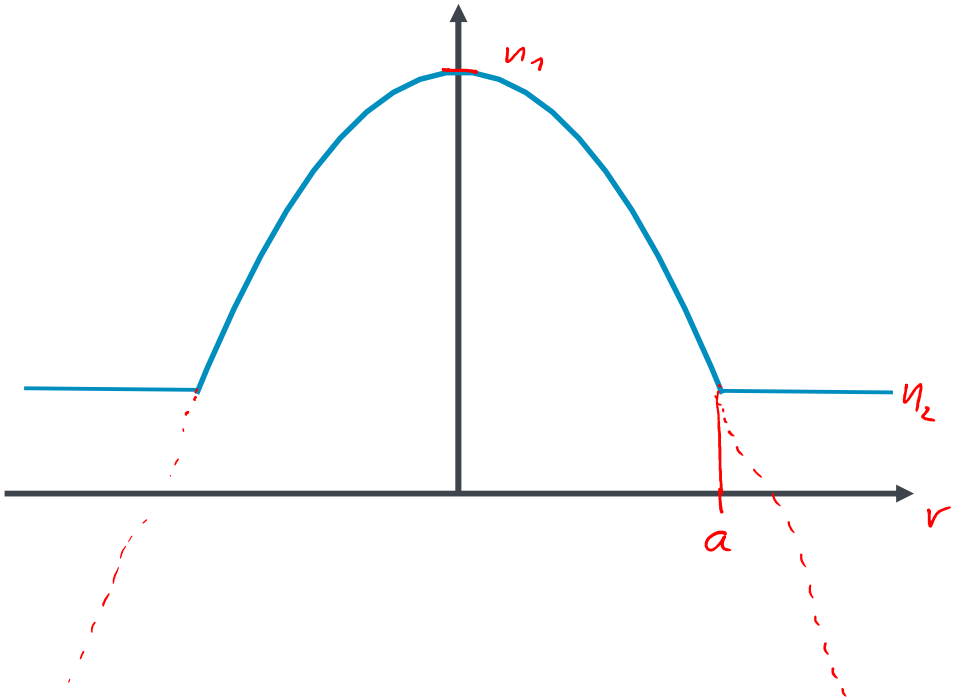
University of Stuttgart
Germany

$$\Delta t \approx \frac{N_1 L}{2c \cdot n_1^2} NA^2$$





Graded-index fibers



$$n^2(r) = \begin{cases} n_1^2\left(1 - 2\Delta\left(\frac{r}{a}\right)^{\alpha}\right) & r \le a \\ n_2^2 & r > a \end{cases}$$

Graded-index fibers

University of Stuttgart
Germany

$$\nabla^2 \underline{E}_{x,y} + k_0^2 n^2(r)\, \underline{E}_{x,y} = 0$$

$$\underline{E}_x = \psi(r,\varphi) \exp(-j\beta z)$$

$$\nabla_t^2 \psi + (k_0^2 n^2(r) - \beta^2)\, \psi = 0$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r \cdot \frac{\partial \psi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \psi}{\partial \varphi^2} + (k_0^2 n^2(r) - \beta^2)\, \psi = 0$$

Graded-index fibers

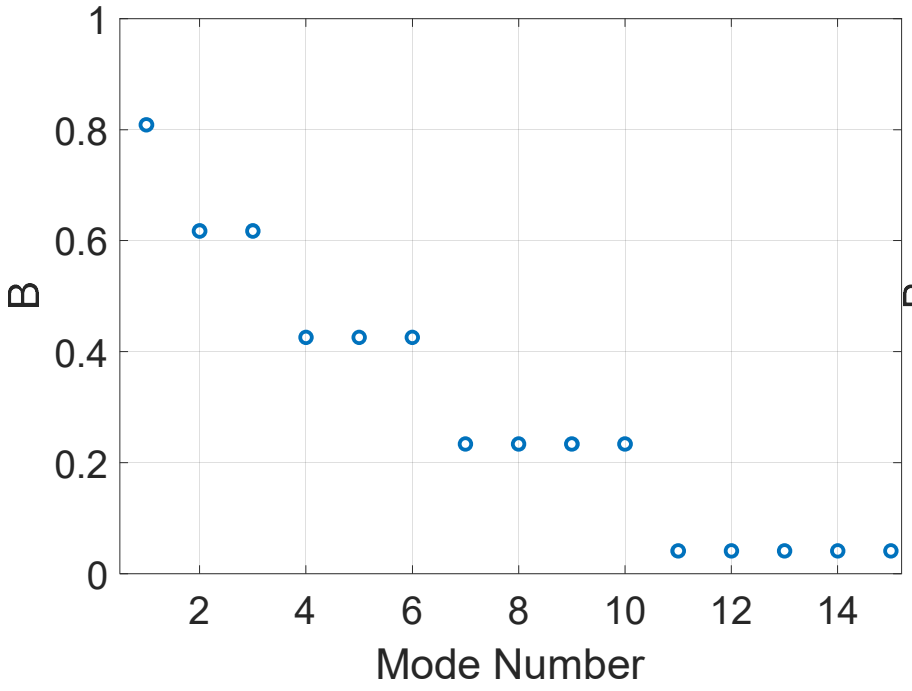
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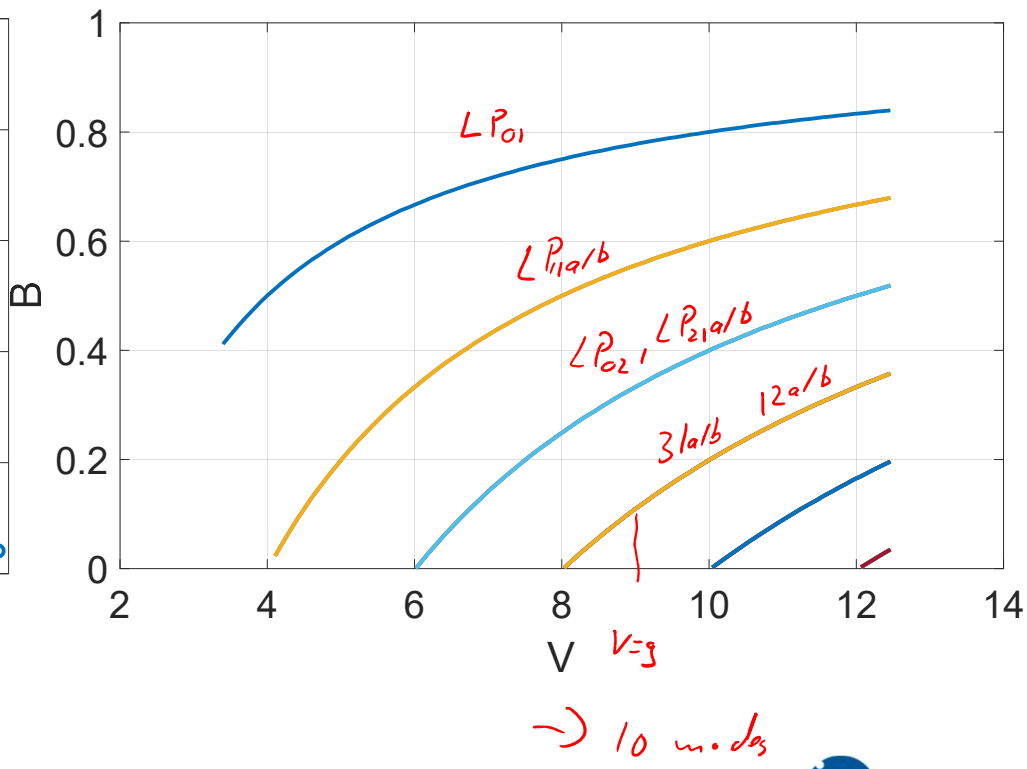
$$E_{x,l,p}(r,\varphi) = C'_{lp}\left(\frac{r}{\xi}\right)^{l} L_{p-1}^{(l)}\left(\frac{r^2}{\xi^2}\right)\exp\left(\frac{-r^2}{2\xi^2}\right)\begin{cases}\cos(l\cdot\varphi)\\ \sin(l\cdot\varphi)\end{cases}$$

$L_{p-1}^{(l)}$: Laguerre Polynomial order $p-1$ and l

$$\beta_{lp} = n_1 k_0\left(1 - \frac{\sqrt{2\Delta}}{n_1 k_0 a}(2l+4p-2)\right)^{1/2}$$

Graded-index fibers







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Germany

Concept Test

Concept test

University of Stuttgart
Germany

For V = 9, which fiber supports more propagating modes?

A Graded index fiber

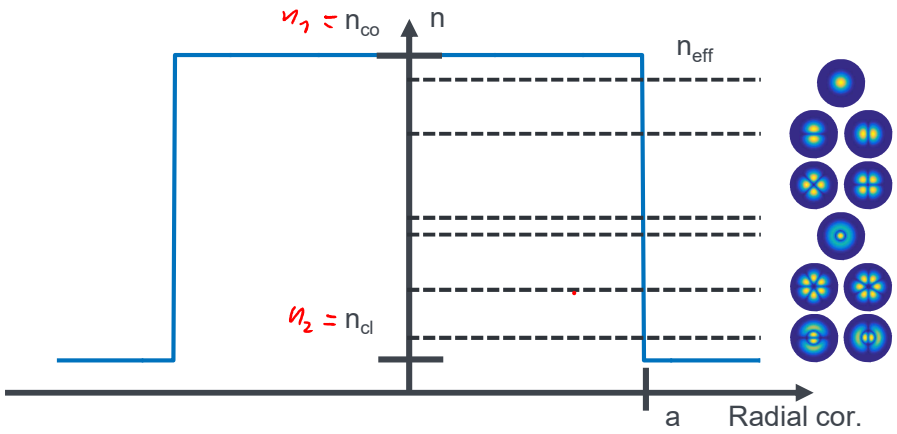
B Step index fiber

C Both have the same

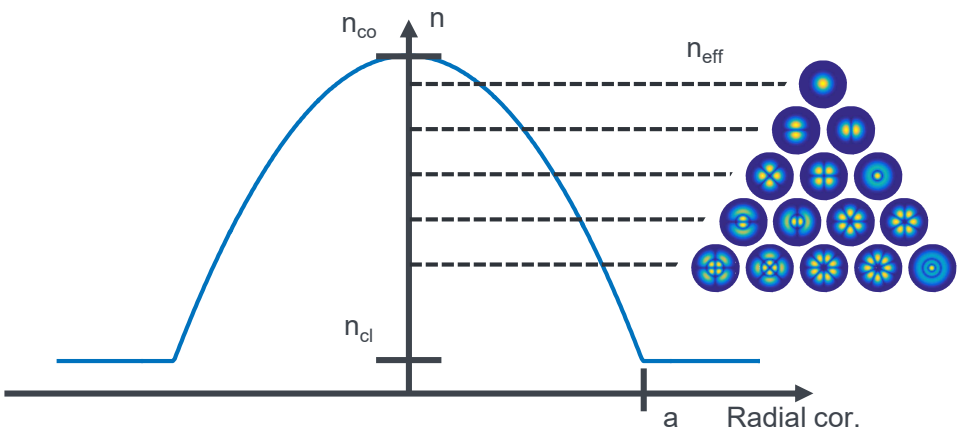
https://ilias3.uni-stuttgart.de/vote/**VGN9**

Multi-mode fibers (MMFs)

University of Stuttgart
Germany



- More Modes
- easy to make



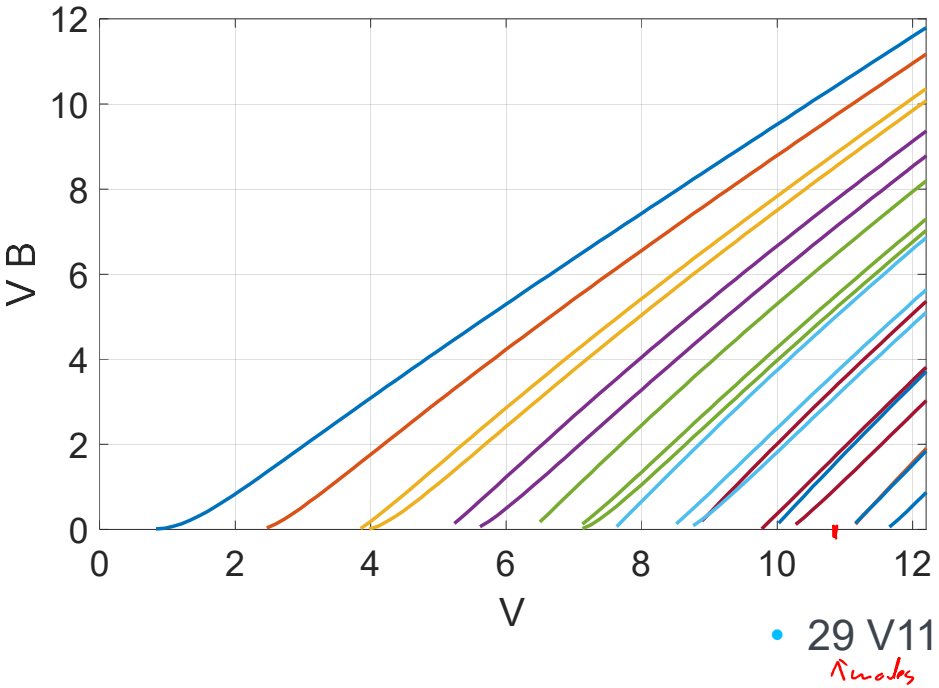
- low DMD
- Structure of mode groups

Graded-index fibers

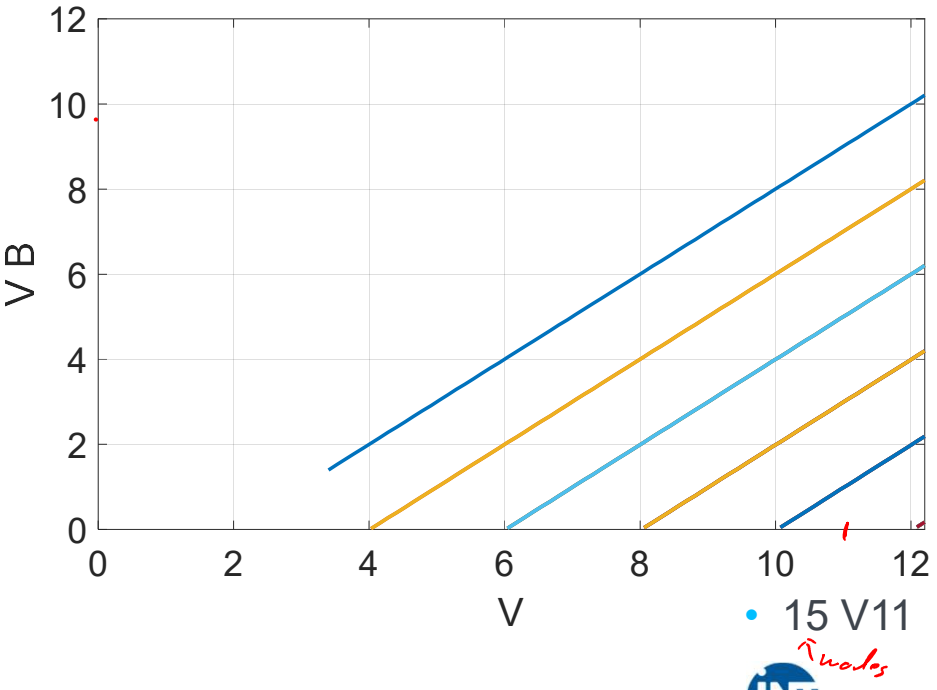
$$\tau = \frac{n_1 - n_2}{c} \cdot \frac{d(V \cdot B)}{dV} + \frac{1}{c} N_2$$

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Step-index

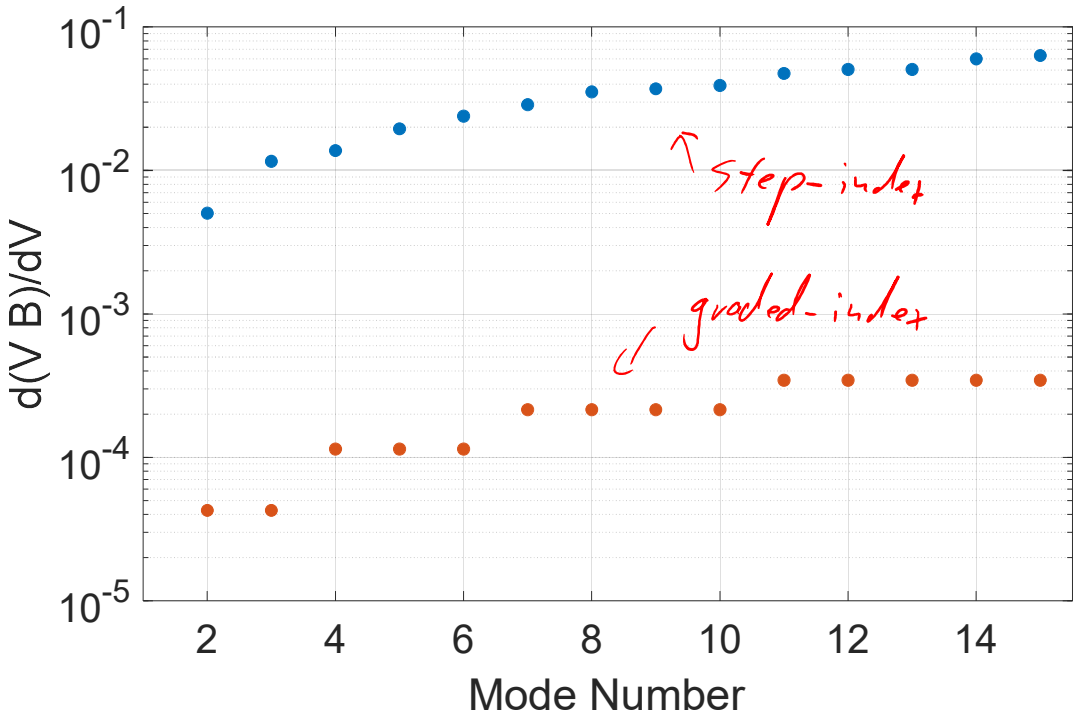


graded-index

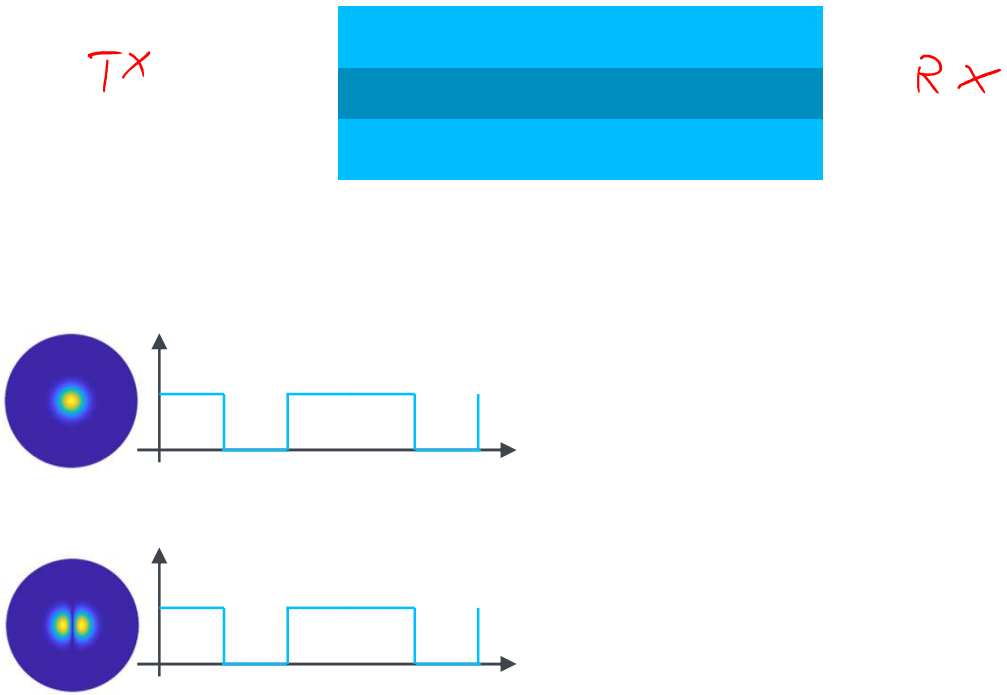


Graded-index fibers

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Germany



Why is the delay difference so important?



Why is the delay difference so important?

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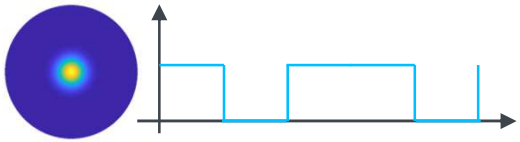


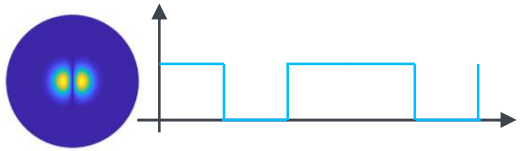


Why is the delay difference so important?

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Why is the delay difference so important?

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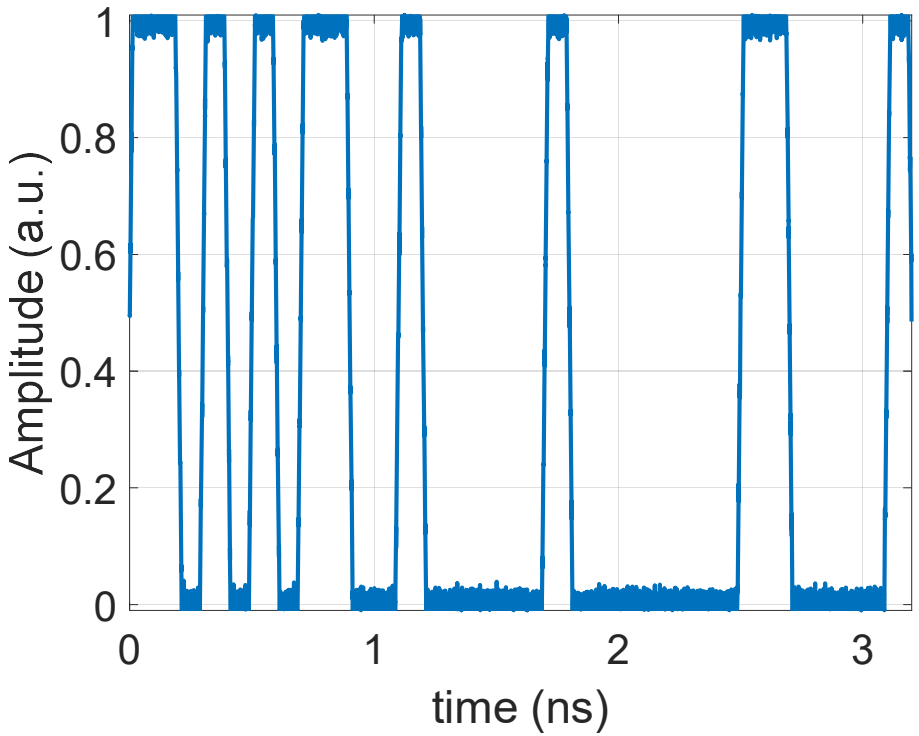


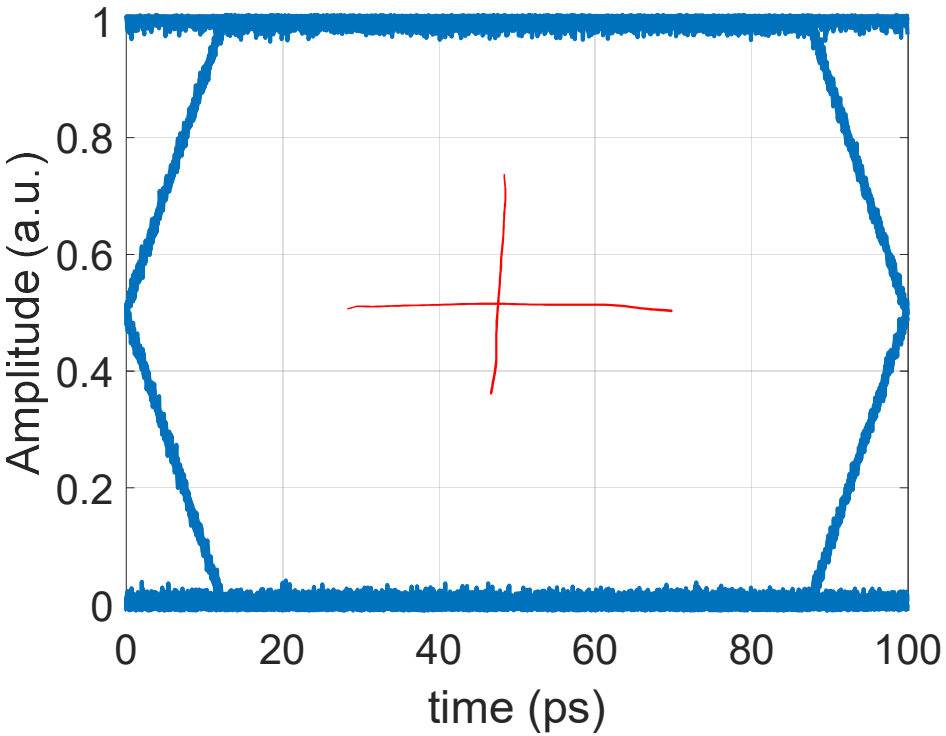


Eye-Diagram

Symbol-rate : 10 GBaud

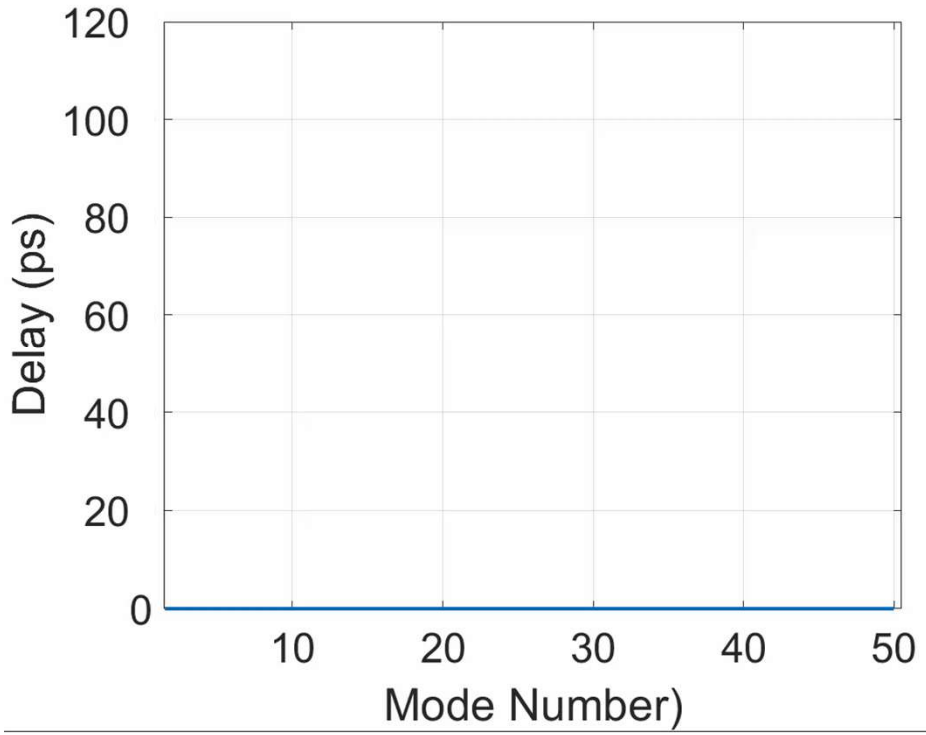
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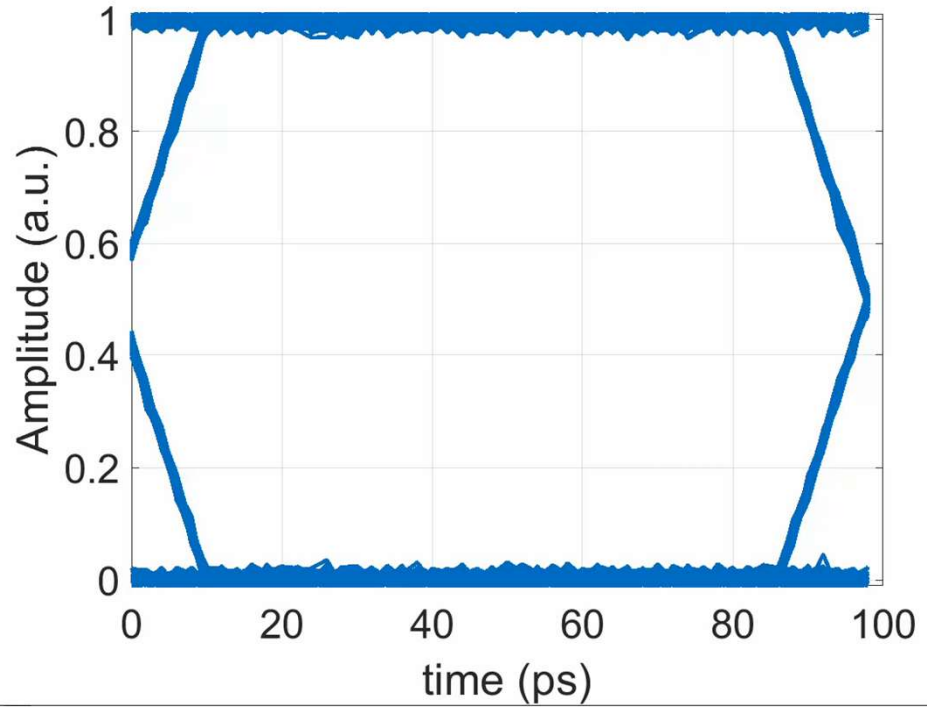




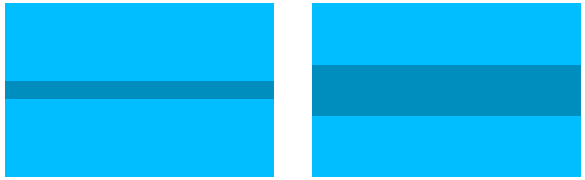
Why is the delay difference so important?

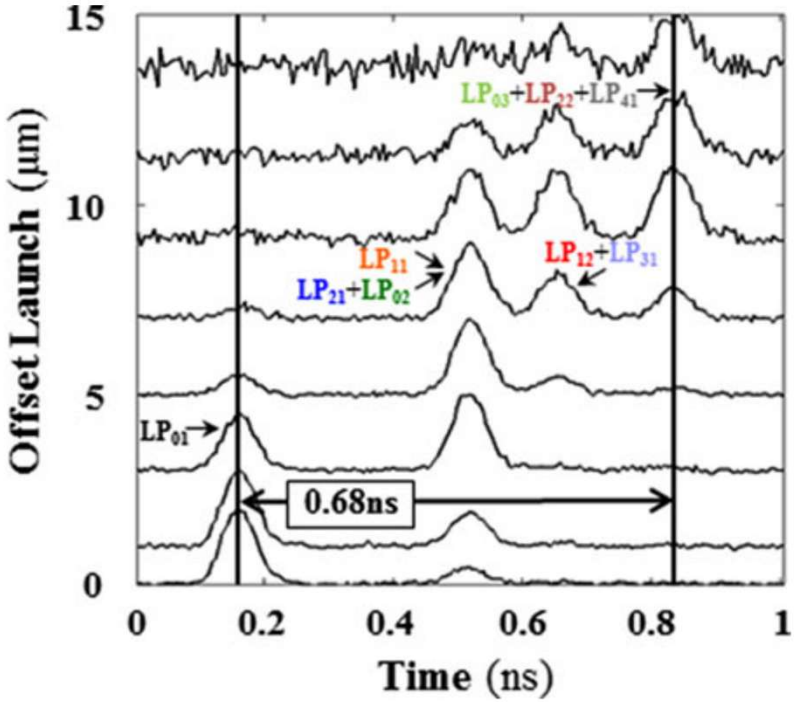
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Graded-index fibers





4.4 km, 15 modes

P. Sillard et al., Low-differential-mode-group-delay 9-LP-mode fiber, JLT 2016



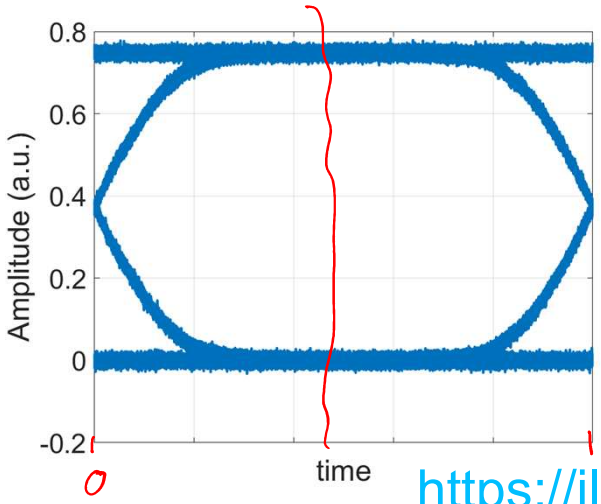
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Germany

Concept Test

Concept test

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Germany

How large can the differential mode-group-delay get so that the eye diagram is still open? Consider a Symbolrate of 50 Gbd, 10 km fiber length, and two modes.



A 20 ps/km

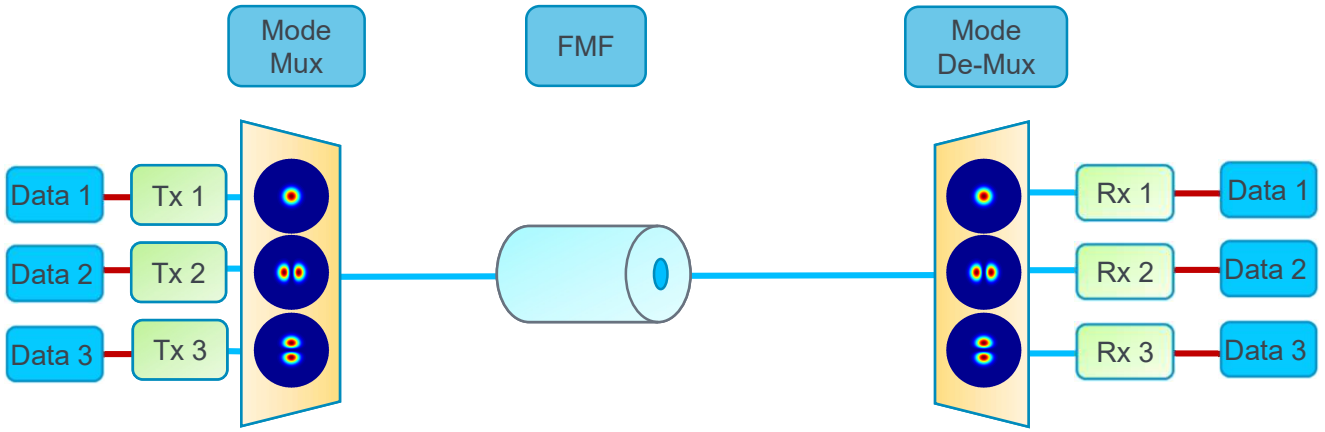
B 0.5 ps/km

C 1 ps/km

https://ilias3.uni-stuttgart.de/vote/**W4IQ**

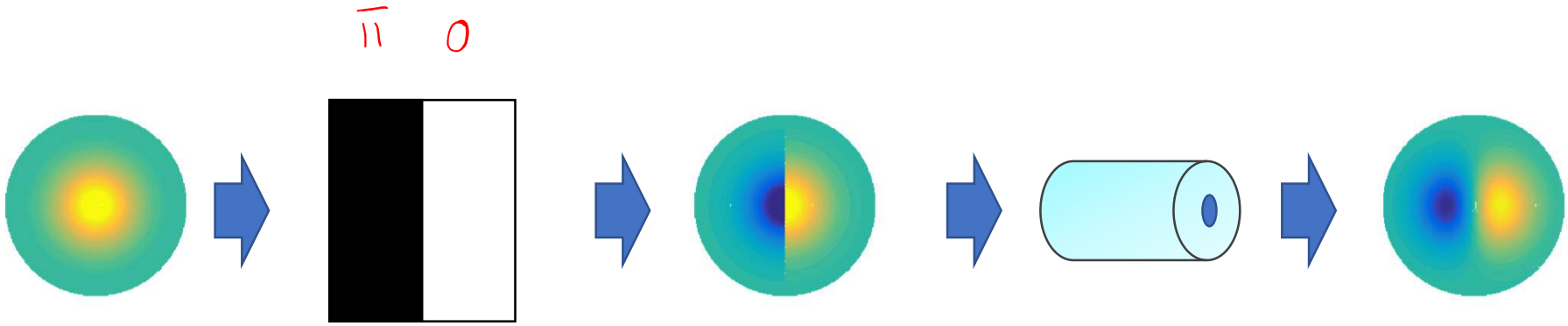
Space division multiplex with MMF

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Germany



Mode-Multiplexers

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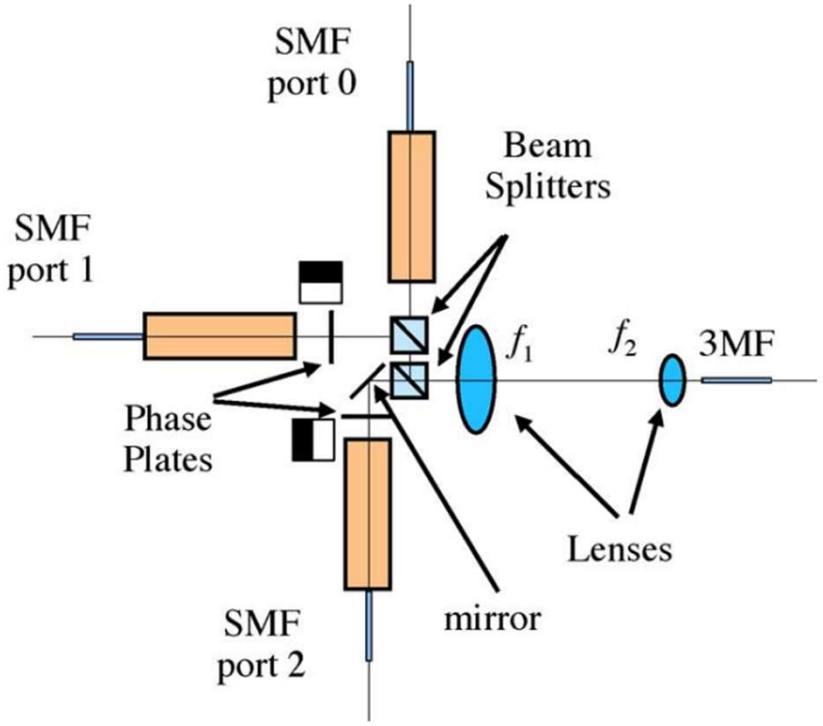
JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 30, NO. 4, FEBRUARY 15, 2012

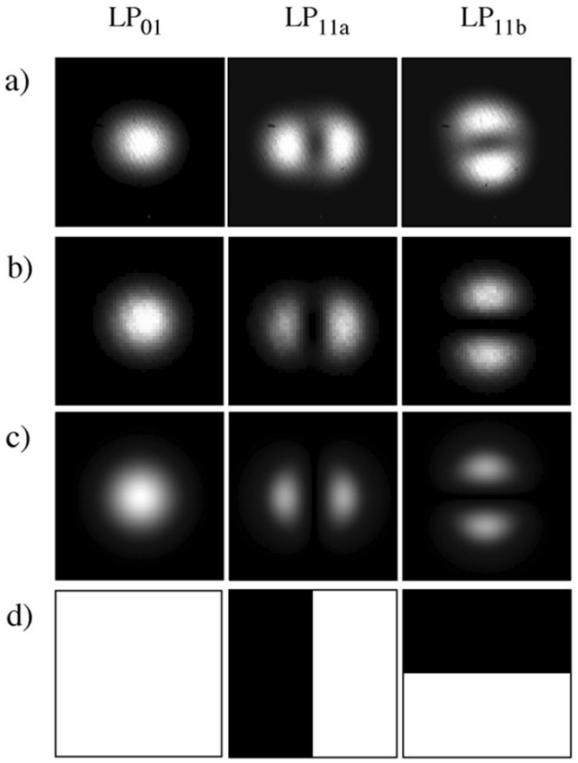
Mode-Division Multiplexing Over 96 km of Few-Mode Fiber Using Coherent 6×6 MIMO Processing

Roland Ryf, *Member, IEEE*, Sebastian Randel, *Senior Member, IEEE*, Alan H. Gnauck, *Fellow, IEEE, Fellow, OSA*, Cristian Bolle, Alberto Sierra, *Student Member, IEEE*, Sami Mumtaz, Mina Esmaeelpour, Ellsworth C. Burrows, René-Jean Essiambre, *Senior Member, IEEE, Fellow, OSA*, Peter J. Winzer, *Fellow, IEEE, Member, OSA*, David W. Peckham, *Senior Member, IEEE*, Alan H. McCurdy, *Member, IEEE*, and Robert Lingle, Jr., *Member, IEEE*

(Invited Tutorial)

Mode-Multiplexers





nature COMMUNICATIONS

ARTICLE

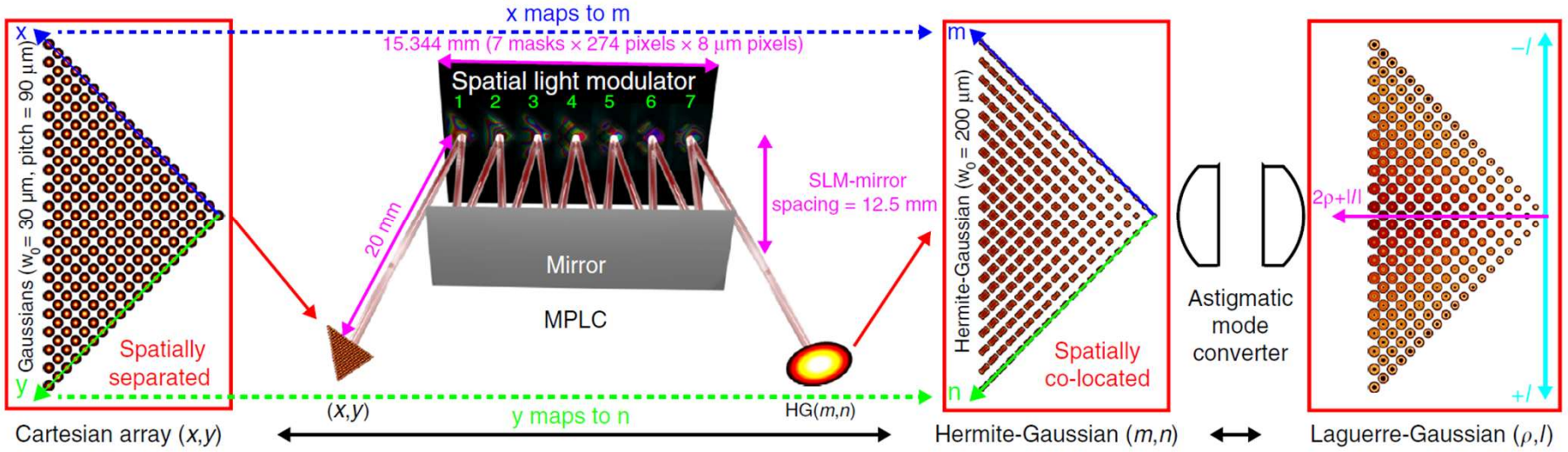
https://doi.org/10.1038/s41467-019-09840-4 OPEN

Laguerre-Gaussian mode sorter

Nicolas K. Fontaine[1,3], Roland Ryf[1], Haoshuo Chen[1], David T. Neilson[1], Kwangwoong Kim[1] & Joel Carpenter[2,3]

Multi-plane light conversion

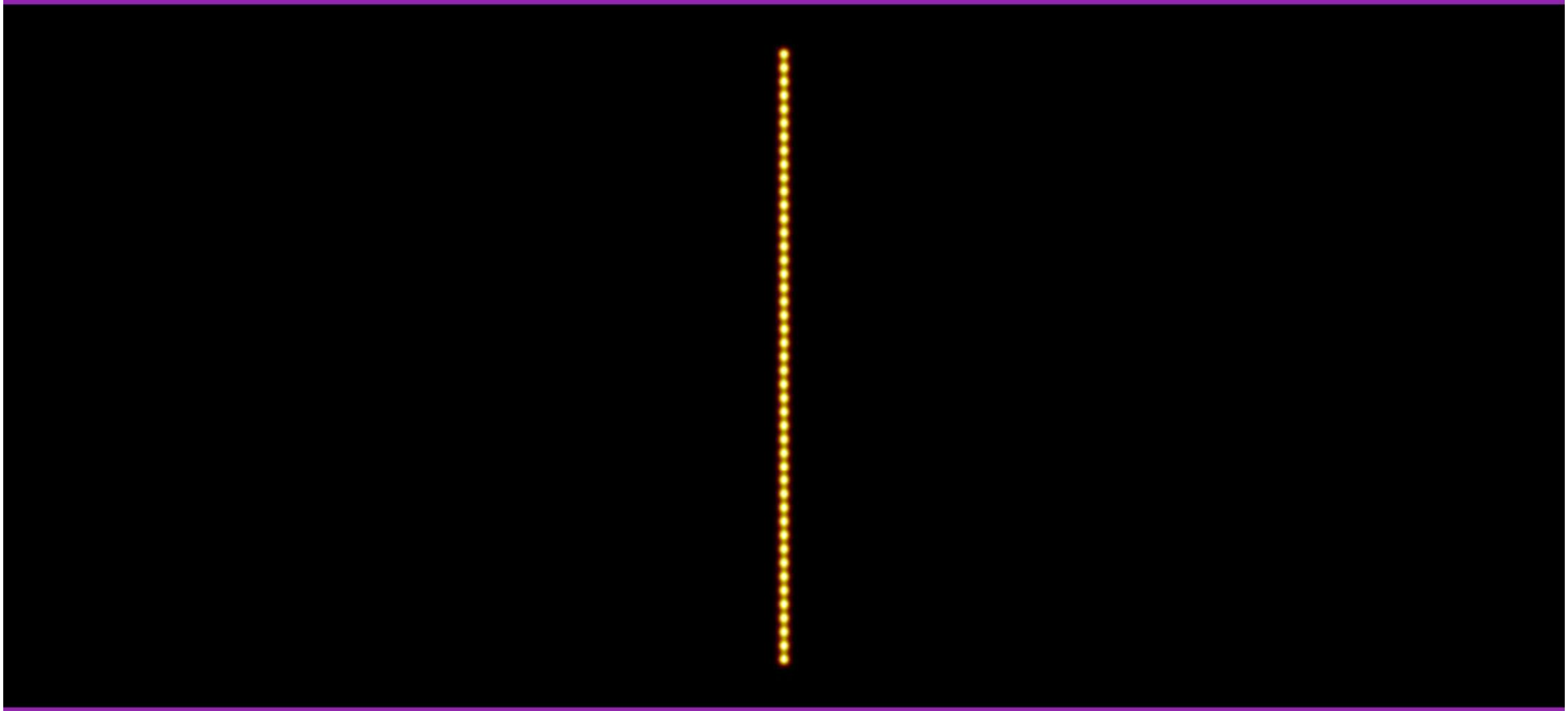
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AUSTRALIA

Multi-plane light conversion

NOKIA Bell Labs

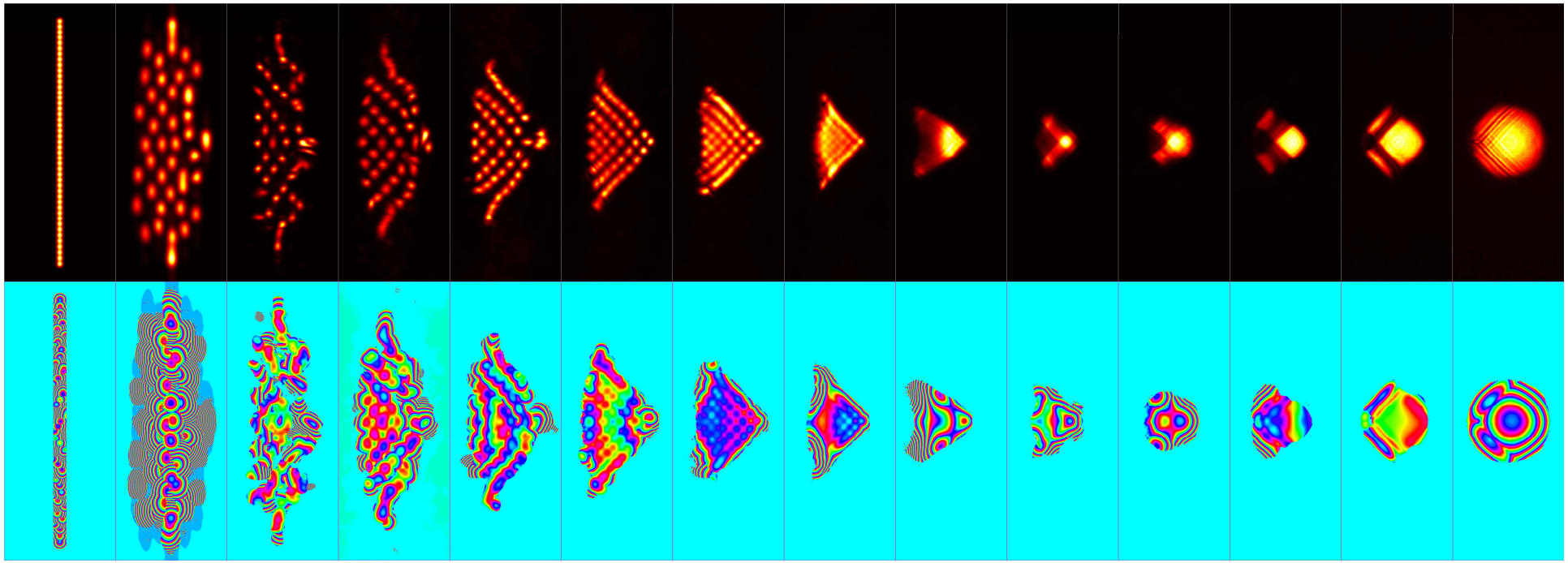


J.-F Morizur et al., J. Opt. Soc. Am. A, 27 (11) 2524-2531 (2010).

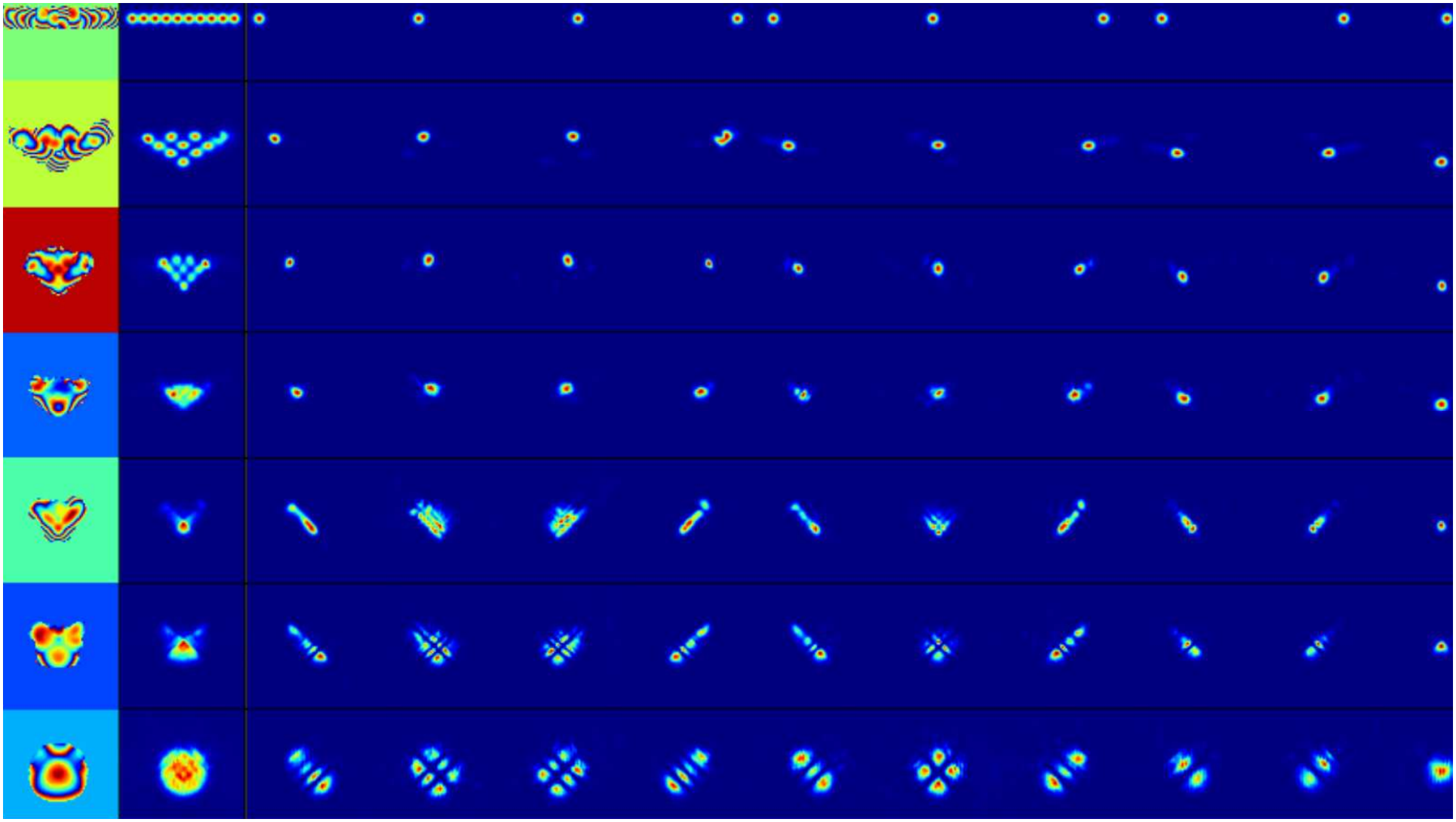
THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Triangular → HG

NOKIA Bell Labs

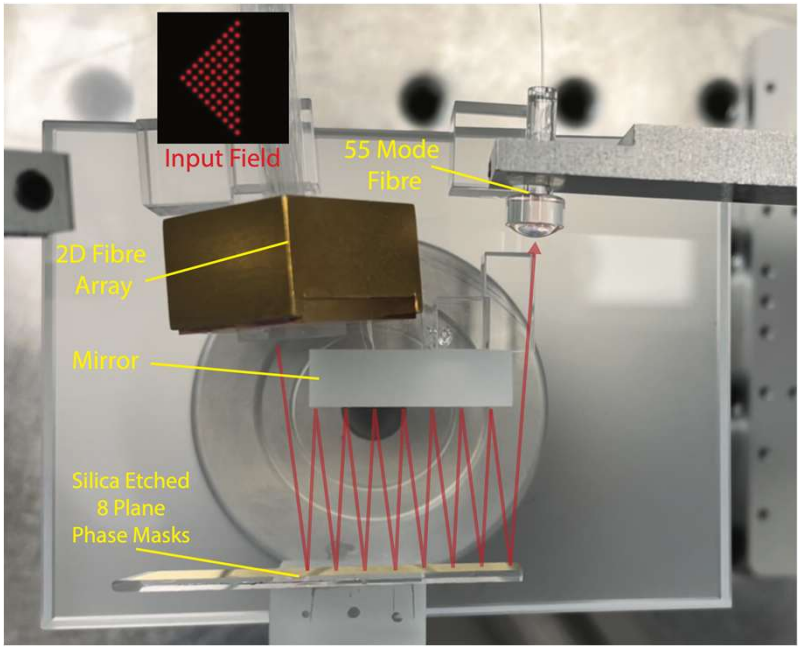


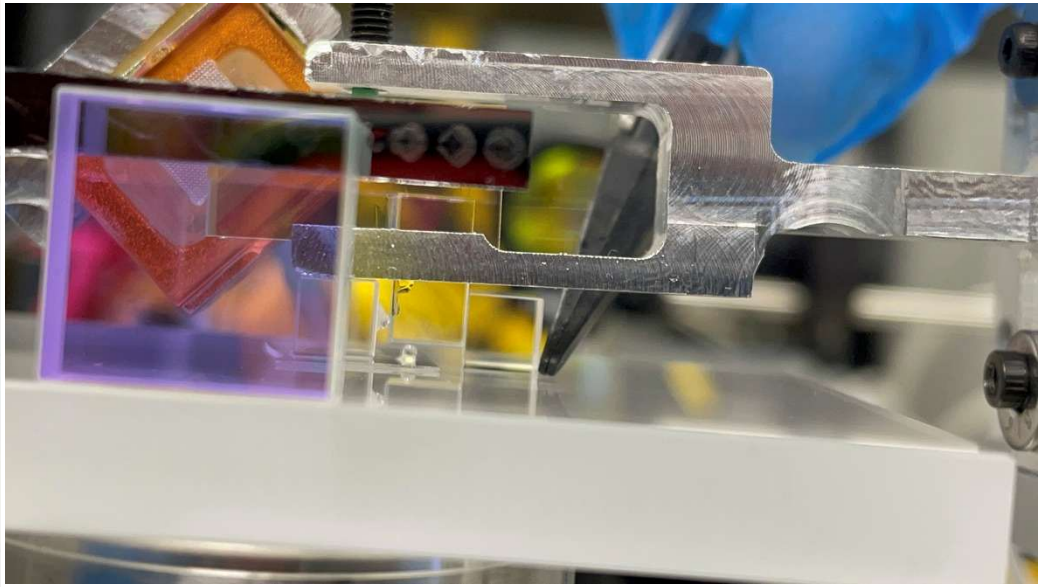
45 modes, 14 planes (512×1280 5μm pixels @ 1550nm, 24mm plane spacing)



Mode-Multiplexers

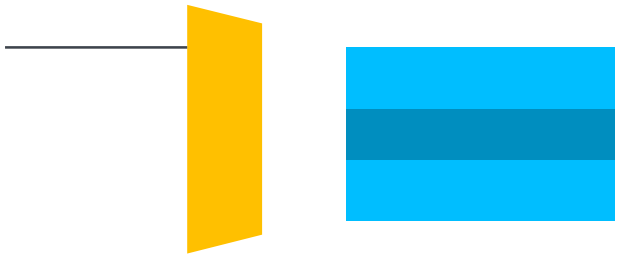
University of Stuttgart
Germany

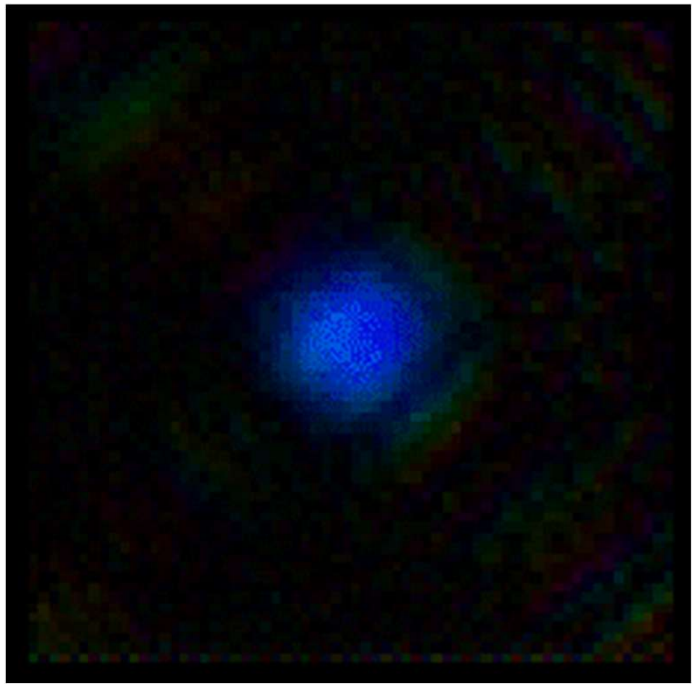




Mode-Multiplexers

University of Stuttgart
Germany





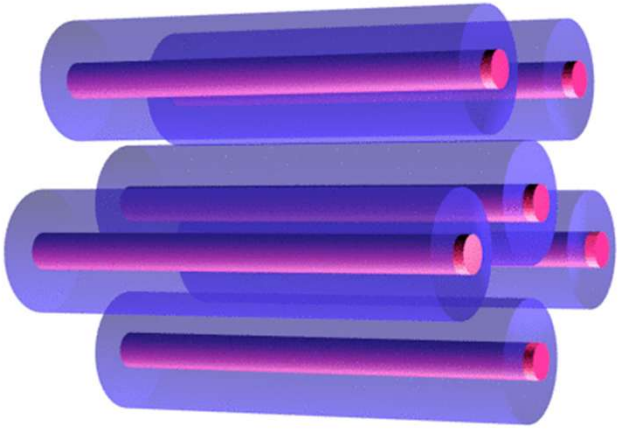


Photonic Lanterns, 3-D Waveguides, Multiplane Light Conversion, and Other Components That Enable Space-Division Multiplexing

This article reviews the mode-shaping technologies needed to perform mode-division multiplexing on few-mode fibers and discusses implementations and tradeoffs of four different techniques: multiplane light conversion, fused fiber devices, 3-D waveguides in glass, and free-space imaging systems.

By NICOLAS K. FONTAINE, *Senior Member IEEE*, JOEL CARPENTER, *Member IEEE*, SIMON GROSS, SERGIO LEON-SAVAL, YONGMIN JUNG, DAVID J. RICHARDSON, *Fellow IEEE*, AND RODRIGO AMEZCUA-CORREA

Mode-Multiplexers: Photonic Lantern



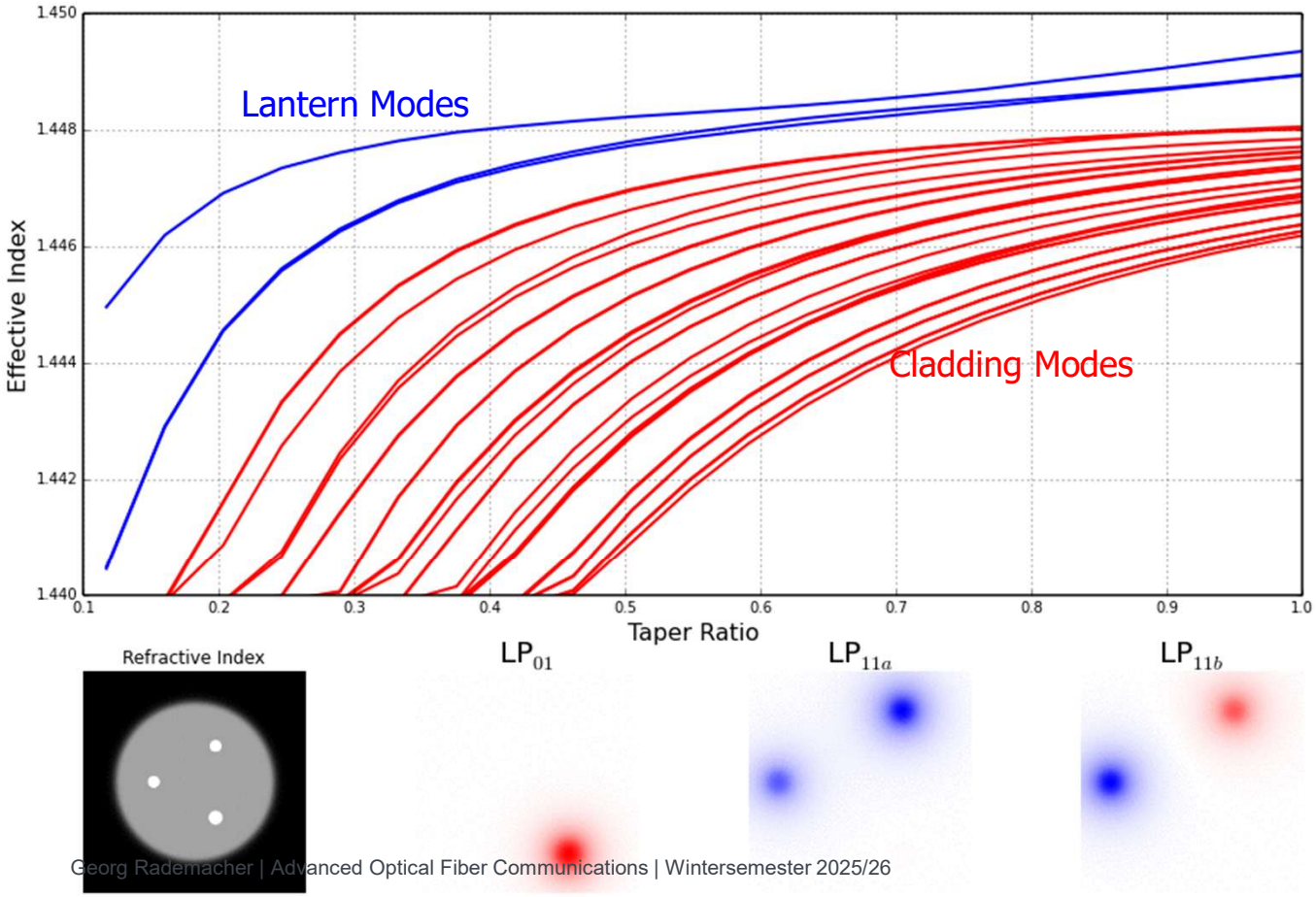


issimilar Fiber Mode Selective Lantern
diabatic transition from groups of cores to modes.

iversity of Stuttgart
rmany

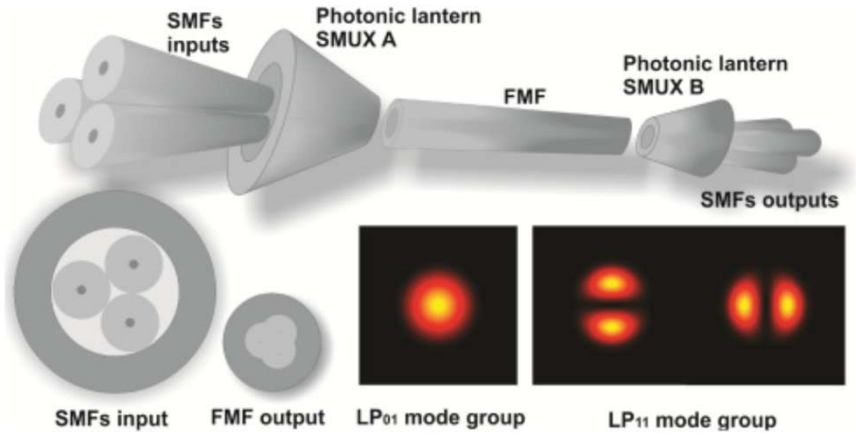
01 Big Core

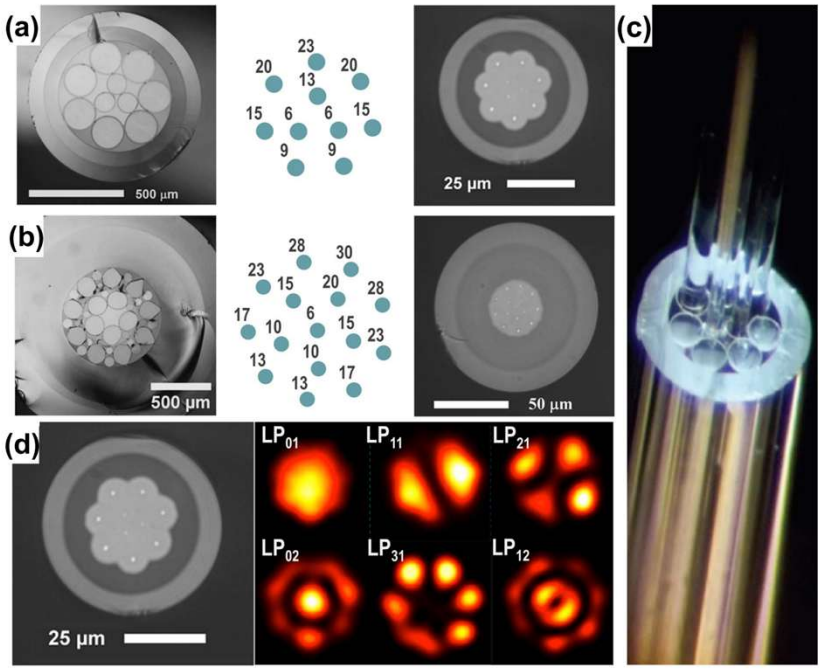
11 Two Smaller Cores

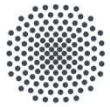


Mode-Multiplexers

University of Stuttgart
Germany







University of Stuttgart
Germany

Nanophotonics 2015; 4:332–352

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Review Article

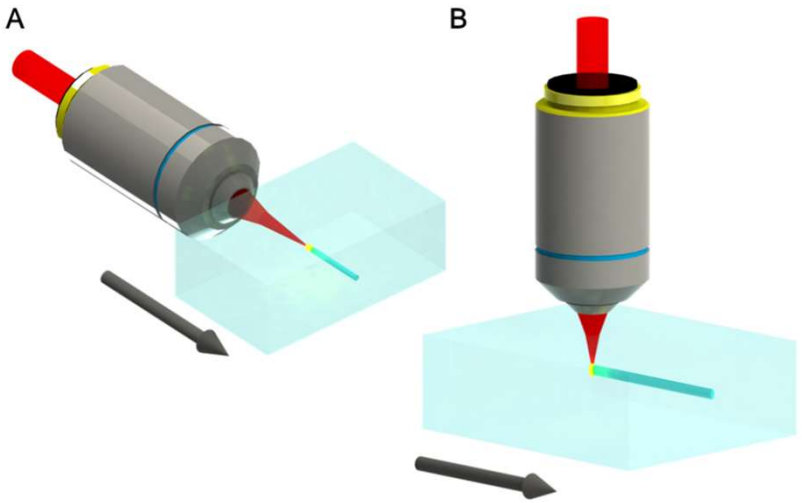
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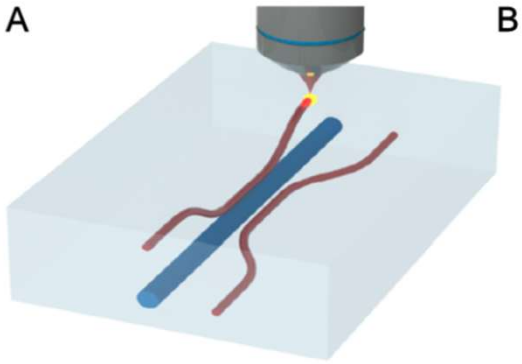
S. Gross* and M. J. Withford

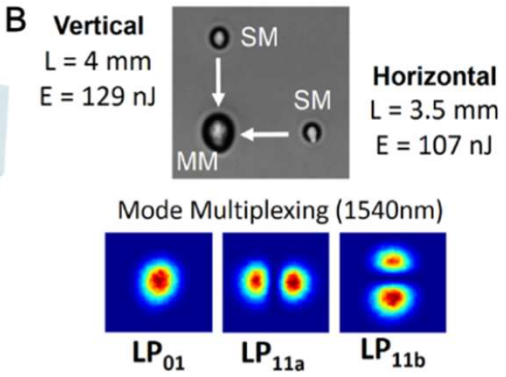
Ultrafast-laser-inscribed 3D integrated photonics: challenges and emerging applications

Mode-Multiplexers

University of Stuttgart
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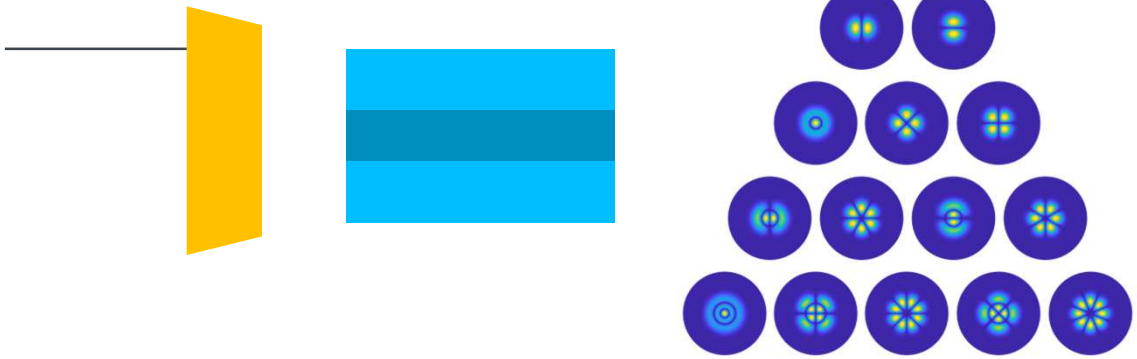


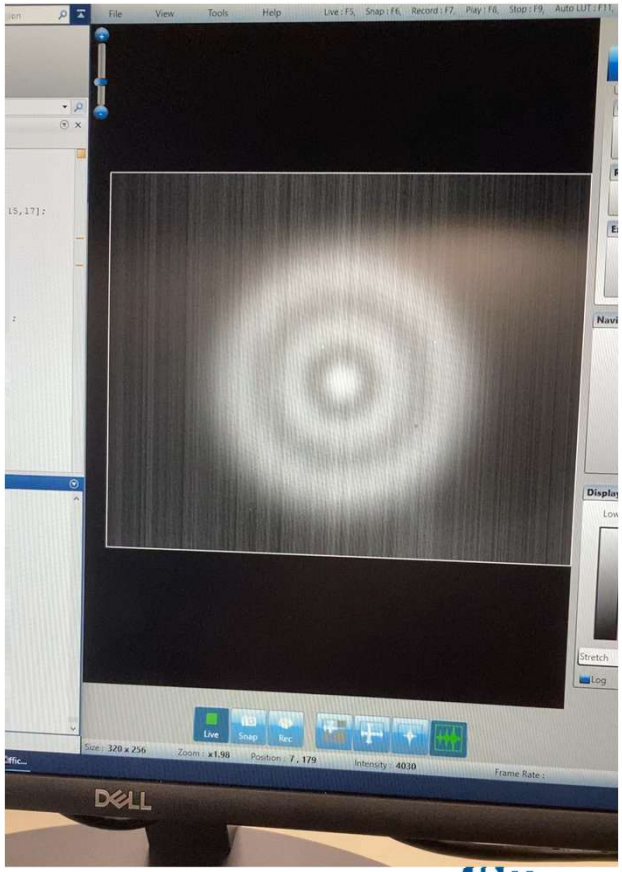


Graded-index fibers

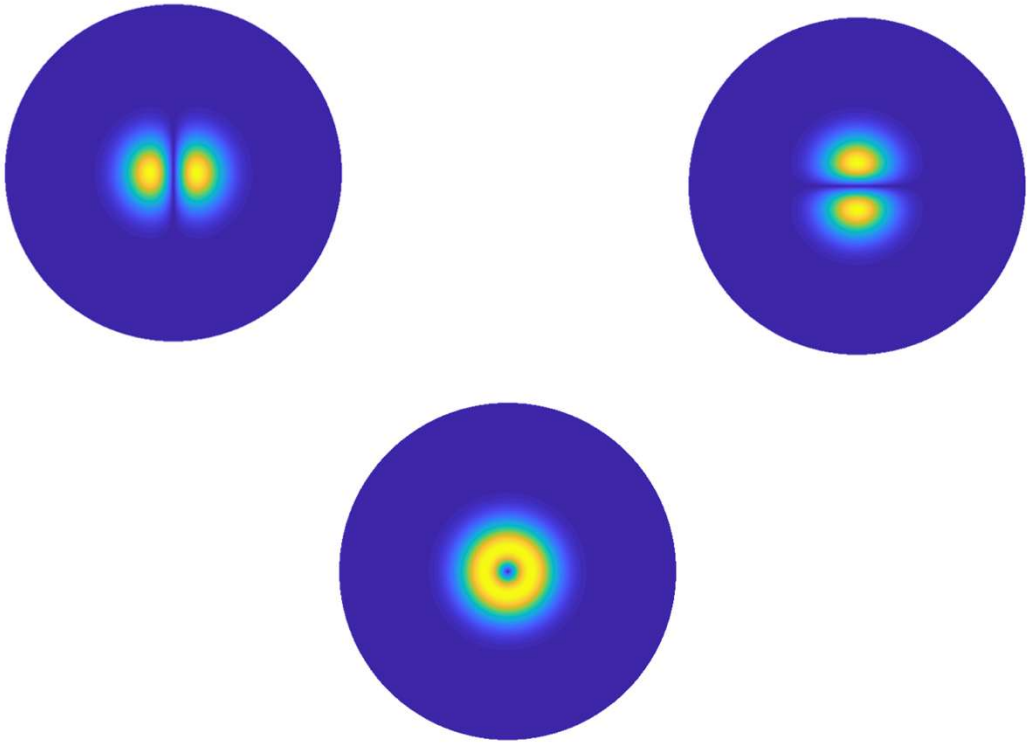
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- SMF
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Mode Coupling



University of Stuttgart
Institute of electrical and optical communications

Thank you!



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